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INFER: Portable hands-on labs for digital forensics education: design, deployment, and educational impact☆

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ABSTRACT

As cybercrime increases annually, digital forensics, a subsection of cybersecurity that involves identifying, preserving, and analyzing digital evidence, has seen a surge in demand over the years. Despite the need, there is a large shortage in the cybersecurity workforce, including digital forensics. Meanwhile, the community also faces a lack of free, interactive, and quality instructional resources for digital forensics education. In this paper, we introduce INFER (INstructional Forensics Education Resource), which contains a set of 18 hands-on labs for digital forensics education. These labs were designed around 12 essential topics, ranging from basic to advanced. They are portable, comprehensive, adjustable, and easy to follow with step-by-step instructions and accompanying lab environments. To evaluate the effectiveness of these labs, we performed a multi-population study involving college students (both graduate and undergraduate), K-12 teachers, and college faculty, using multiple survey instruments. Based on the results, students' self-reported skills and confidence in performing DF tasks significantly improved. We also found that clear, detailed instructions help reduce the lab's difficulty and improve learning effectiveness. Finally, a follow-up survey showed that educators successfully integrated INFER labs into their own curriculum with positive ratings and feedback, reflecting documented classroom adoption by independent educators.

1. Introduction

Digital Forensics (DF) is a subsection of cybersecurity that focuses on identifying, collecting, analyzing, and preserving digital evidence for various digital systems and platforms, including mobile devices, networks, databases, and IoT devices. DF is essential not only for recovering deleted or damaged files from disks and devices but also plays a crucial role in both criminal and civil investigations, (for example, intellectual property disputes, employment matters, and e-discovery), particularly in cases involving cybercrime. According to the FBI's Internet Crime Complaint Center (IC3), cybercrime losses in the United States alone exceeded $12.5 billion in 2023 (Federal Bureau of Investigation, 2023), reflecting the urgent societal need for trained digital forensic professionals capable of investigating and prosecuting such offenses. Furthermore, digital evidence is now present in the majority of criminal cases, making competent DF expertise a cornerstone of modern law enforcement.

Despite growing demand and potential, there is a significant shortage of professionals in the cybersecurity field. Digital forensics, being a specialized subset of cybersecurity, also faces a growing need for experts in this area. In 2024, the global cybersecurity workforce gap is close to 5 million professionals, a 19.1% increase from 2023 (ISC2, 2024). This gap is expected to widen as new digital systems evolve rapidly. As society becomes increasingly reliant on technology and the threat of cyberattacks intensifies annually, the need for skilled cybersecurity professionals to defend against digital threats and investigate the cyber attacks has never been more critical.

Traditional training methods in digital forensics primarily emphasize theory and certifications. However, these approaches may not expose learners to sufficient practical exercises and hands-on experiences. High-quality digital forensic education emphasizes the importance of gaining practical experience with relevant forensic tools and platforms. Experiential learning, extensively studied in STEM fields, has demonstrated a positive impact on student learning outcomes. It is grounded in established pedagogical frameworks, such as Kolb's Experiential Learning Theory (Kolb, 1984), which posits that knowledge is best acquired through concrete experience and reflective practice.

However, its effectiveness has not been thoroughly studied in the field of digital forensics due to the lack of systematic, comprehensive

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hands-on labs that are readily available and easily adaptable for in-depth research. Currently, the most advanced hands-on training in digital forensics remains limited to a small audience.

Several recent digital forensic education projects (UHCL, 2019; NSF, 2015, 2019, 2017) contribute to the field by developing specific courses (inherently a recipe of lectures and labs to fulfill learning outcomes) that cover one or more sub-branches (such as IoT forensics), or by pedagogically enhancing digital forensics education with specific technology (such as Augmented Reality). However, the lack of high quality labs remains a challenge. More specifically, existing labs tend to cover only narrow sub-domains such as disk forensics, while leaving areas such as network forensics, mobile forensics, and IoT forensics underrepresented. In addition, most available labs do not provide reproducible self-contained environments, placing a significant preparation burden on instructors. Critically, few existing efforts have paired their lab designs with rigorous empirical studies to measure their actual impact on student learning. In addition, the impact of experiential learning on digital forensics education is still unclear.

Therefore, this document presents a comprehensive instructional resource for digital forensic education and studies the impact of experiential learning based on a pool of newly developed hands-on labs. Specifically, this work develops an ***INstructional Forensics Education Resource (INFER)*** in the form of step-by-step hands-on labs with accompanying instructional operating environments, and investigates the impact of the labs on student learning in digital forensics. The work in the paper answers the research questions: *To what extent will the INFER labs impact student knowledge, skills, and attitudes regarding digital forensics? How do educators perceive the INFER labs in terms of usability and difficulty? To what extent do educators adopt and integrate INFER hands-on labs into their own classrooms?* More specifically, what is the impact on the students' self-reported learning gains, perceived skills, interest in forensics, and career readiness? Our evaluation, conducted in multiple cohorts of undergraduate and graduate students, shows significant gains in students' self-reported forensic knowledge, perceived practical skills, and interest in pursuing careers in the DF field.

In this work, the designed INFER labs come with the required instructional operating environments, so instructors do not have to prepare them from scratch. The INFER labs are portable, comprehensive, expandable, and adjustable: (1) they systematically and comprehensively cover a number of forensics sub-areas in depth; (2) they are ready-to-use with the required lab environments and materials; and (3) they can be adopted for different levels of digital forensics education, such as undergraduate and graduate level courses. Like SEED Labs (Du and Wang, 2008), INFER can be adopted with proper adjustment by other courses in high school education and workforce development, such as summer academies or summer camps (e.g. GenCyber (GenCyber, 2025)); community college courses; and training for digital forensic practitioners, law enforcement personnel, and IT professionals from nonprofit organizations and government agencies. Finally, this paper itself is an extension of the cumulative work published in the IFIP WISE 2025 conference (Huynh et al., 2025a) by mapping INFER labs to the NICE Cybersecurity Workforce Framework, adding faculty follow-up classroom-adoption survey, and correcting duplicate survey responses in the earlier results.

2. Background and related work

2.1. Approaches in cybersecurity education

There are several approaches to teaching and introducing cybersecurity concepts to new learners, including courses, workshops, games, and cyber ranges. Traditional methods often rely on lectures focused on theoretical knowledge and the concepts of cybersecurity (Friedl et al., 2022). Cyber ranges are valuable tools for simulating real-time cyber scenarios (Katsantonis et al., 2023; Leitner et al., 2021; Oh et al., 2020); however, they can be costly and often complex and challenging for beginners. Hands-on labs (Du and Wang, 2008), workshops (Yuan et al., 2014; Englbrecht and Pernul, 2020), and serious games (Friedl et al., 2022, 2024; Pan et al., 2015) are effective tools to engage new learners. These methods allow learners to gain practical experience through real-world use cases, fostering deeper engagement with the material, skill development, and a better understanding of core concepts. Although games offer interactive and visually appealing experiences, they often lack the depth necessary to build proficiency with the tools used in the field. For all of the methods mentioned above, most of them are for general cybersecurity education and are not directly designed for digital forensics education. This subdomain requires familiarity with specific tools such as Autopsy and Volatility. In addition, practitioners must follow strict investigative workflows and legal frameworks while working across various platforms, including computer file systems, memory, networks, and even mobile devices.

2.2. Current approaches in digital forensics education

Given these unique challenges, several efforts have been made to develop DF education training. However, there remains a notable lack of comprehensive curricular and teaching materials for digital forensics, particularly at the undergraduate and postgraduate levels. Although some work has been done to enhance digital forensic education through hands-on labs and activities (Chi et al., 2009; Ward, 2021; Xu et al., 2022; Wang et al., 2024), these efforts have various limitations. Chi et al. (2009) focused primarily on disk-related topics and faced significant hardware and software cost barriers. Ward (2021) included digital forensics as part of a broader cybersecurity program, but covered only a narrow set of topics and required dedicated IT infrastructure to deploy labs. Xu et al. (2022) offered a customized virtual machine and scenario-based case studies across multiple forensic domains, but relied on presentation slides rather than structured step-by-step instructions and lacked formal evaluation of student outcomes. Wang et al. (2024) designed categorized hands-on activities with positive student feedback, but provided no pre-configured environment and no systematic pre- and post-assessment. Overall, as summarized in Table 1, none of these efforts fully addresses topic breadth, tool diversity, environment availability, instructional guidance, and empirical evaluation simultaneously, which motivates the design of INFER.

2.3. Challenges in digital forensics education

The limitations identified in existing digital forensics educational efforts reflect a set of deeper, structural challenges that make this subdomain particularly difficult to teach effectively. These challenges are interconnected: the breadth and complexity of the field drives high environment setup costs, which in turn discourages the development of comprehensive instructional materials, ultimately leaving the impact of experiential learning in this domain poorly understood.

To begin with, digital forensics is inherently a multi-tool, multi-platform discipline that spans a wide range of technical domains. Practitioners must be proficient in specialized tools such as Autopsy for disk analysis, Volatility for memory forensics, Wireshark for network traffic inspection, and Ghidra for reverse engineering, among others. Beyond tool proficiency, investigations must be conducted across diverse platforms including Windows and Linux file systems, Android mobile devices, network infrastructure, and emerging environments such as IoT and drone systems. As noted by Humphries et al. (2021), current training offerings in specific subfields such as mobile forensics tend to cover only acquisition and analysis, often failing to address the full investigative pipeline. This multi-tool, multi-platform breadth makes it difficult to design educational materials that are both comprehensive and accessible within the constraints of a single course.

This complexity, in turn, drives a second and closely related challenge: the significant overhead required to prepare functional lab environments. Meaningful digital forensics exercises depend not only on

Table 1
Comparison of existing digital forensics education efforts.

Work	Breadth	Tool diversity	Environment	Step-by-Step	Evaluation
Chi et al. (2009)	Low	Low	✗	Partial	✗
Ward (2021)	Low	Low	✗	Partial	✗
Xu et al. (2022)	Med	Med	Partial	Partial	✗
Wang et al. (2024)	Med	Med	✗	✓	Partial
INFER (Ours)	**High**	**High**	✓	✓	✓

correct software configurations but also on carefully constructed artifact files, including disk images, memory dumps, network capture files, and multimedia evidence. As Flores et al. (2021) report, both students and educators highlight challenges during the hands-on phases of digital forensics training, emphasizing the need for structured guidance and better instructional support to facilitate effective learning. Ward (2021) similarly noted that deploying lab exercises required significant coordination with campus IT infrastructure to establish properly isolated environments. When such support is unavailable, instructors must build these environments from scratch — a process that is time-consuming and often inconsistent across institutions.

A third challenge is the legal and procedural dimension that is unique to digital forensics and largely absent from general cybersecurity education. Forensic practitioners must adhere to strict evidence handling protocols, including maintaining an unbroken chain of custody, to ensure that evidence remains admissible in legal proceedings. These procedural requirements demand not just technical proficiency but a professional mindset that is difficult to develop through lectures or game-based activities alone. Existing educational resources rarely address this dimension explicitly, creating a gap between academic training and the standards expected in professional and legal practice.

Finally, the broader field lacks a sufficient empirical basis for understanding how well different instructional approaches actually work. A recent systematic review of digital forensics education found that existing programs are fragmented and inconsistently structured, often lacking standardized curricula and measurable learning outcomes (Stigall and Choo, 2021). While Kolb's Experiential Learning Theory (Kolb, 1984) provides a well-established pedagogical foundation, arguing that knowledge is best acquired through concrete experience and reflection, its applicability and effectiveness in the specific context of digital forensics has not been thoroughly validated. This absence of rigorous empirical evaluation makes it difficult for educators to make informed decisions about curriculum design and instructional strategy. Addressing these interconnected challenges requires not only comprehensive and ready-to-use instructional materials, but also a commitment to systematically measuring their impact across diverse learner populations, which motivates the design of INFER to address these challenges.

3. INFER — hands-on labs for forensic education

3.1. Motivation

The lack of quality hands-on labs makes many digital forensics courses impractical and superficial. For example, students completing a course may understand the concept of data acquisition (acquiring evidence), but they may not know how to actually zero-out (completely wipe) a target disk, construct a brand-new file system on it, and then perform data acquisition in a Linux environment. Similarly, they may have learned the concept of file carving (recovering files from unallocated areas), but they may not know how to manually locate file fragments on the disk and reassemble them to recover the file. Hence, well-designed hands-on labs will be indispensable for helping students delve into digital forensics and gain practical experience. In addition, the delivery of these labs can help strengthen our understanding of the distinctive impact of experiential learning on the education of digital forensics.

Therefore, INFER labs are designed to address gaps in current digital forensics education by being portable, comprehensive, expandable, adaptable, and ready to use by anyone with the lab environments and materials provided. By creating a structured learning framework, we can effectively convey advanced concepts to new learners while keeping them engaged through practical, real-world applications and exercises.

3.2. Lab design and curriculum alignment

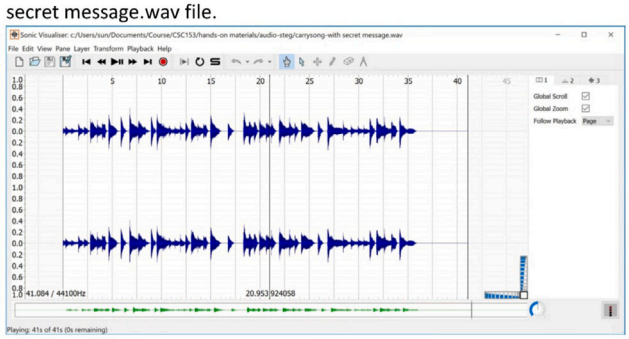
Our previous work (Huynh et al., 2025b) presented the INFER lab framework, designed based on Kolb's Experiential Learning Cycle, and its alignment with Knowledge Units (KUs) (National Security Agency (NSA), 2024) and CAE-CD guidelines (National Security Agency (NSA)) to support cybersecurity education. Although this mapping supports curriculum design and academic learning objectives, it does not address workforce-oriented skills and knowledge requirements. In this work, we extend the framework by mapping INFER hands-on labs to the NICE (National Initiative for Cybersecurity Education) Cybersecurity Workforce Framework (National Initiative for Cybersecurity Careers and Studies, 2026). NICE is a workforce framework built on Task and KSA (Knowledge, Skills, and Abilities) statements. Unlike CAE-CD, which was built for academic learning objectives, NICE organizes around work roles and job competencies. The mapping between NICE and INFER labs is many-to-many. A single lab addresses multiple KSAs, and a single KSA can also appear across multiple labs and topics. Table 5 in the Appendix shows this mapping and the many-to-many relationship. For example, Skill *S0607 — Collecting digital evidence* appears across "Android Forensics" and "Drone Forensics" hands-on. Note that not all KSAs are topic-specific; many are general foundational competencies applied broadly across different forensic topics, and therefore, are excluded from the table. For example, *S0856 — Digital evidence analysis* and *S0854 — Data analysis* are general skills required across almost all forensics tasks, or *K0696 — Forensic data principles*, *K0676 — Cybersecurity laws and regulations*, and *K0677 — Cybersecurity policies and procedures* are foundational knowledge that must be learned as a prerequisite before students start with the lab exercises. Overall, INFER covered multiple KSAs that align with two primary NICE work roles: Digital Forensics (under the Protection and Defense work role category) and Digital Evidence Analysis (under the Investigation work role category), preparing students for both career pathways in the cybersecurity workforce.

3.3. Lab structure and contents

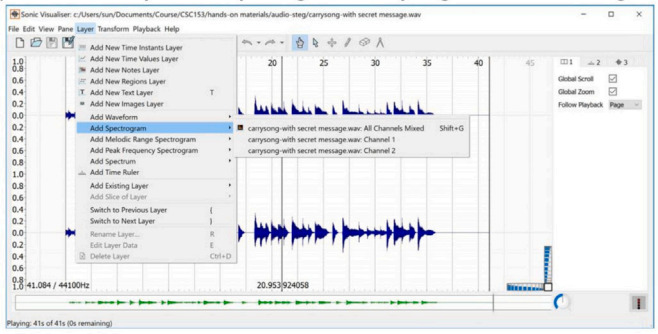
Our work contains 18 step-by-step hands-on exercises that cover 12 essential topics in digital forensics. These topics are categorized and range from foundational concepts, such as data acquisition, to advanced and contemporary areas, including social network forensics, memory forensics, and reverse engineering. Basic labs are estimated to take about an hour. Intermediate problems will require approximately two to four hours to complete, whereas advanced activities will take four to more hours. Most labs can be divided into multiple modules to reduce complexity and allow students to work on a smaller task at a time. A complete list of the 12 topics and 18 labs, along with their mappings to NICE skills and knowledge, is provided in Table 5 (Appendix 7).

As exemplified by Fig. 1, each lab includes detailed step-by-step instructions with corresponding screenshots so that students can work

Step 21: Start Sonic Visualiser program. Click on **File->Open** to open the carrysong-with secret message.wav file.



Step 22: Click on **Layer->Add Spectrogram-> carrysong-with secret message.wav: All Channels Mixed**.



Step 23: You can move around to display the secret message.

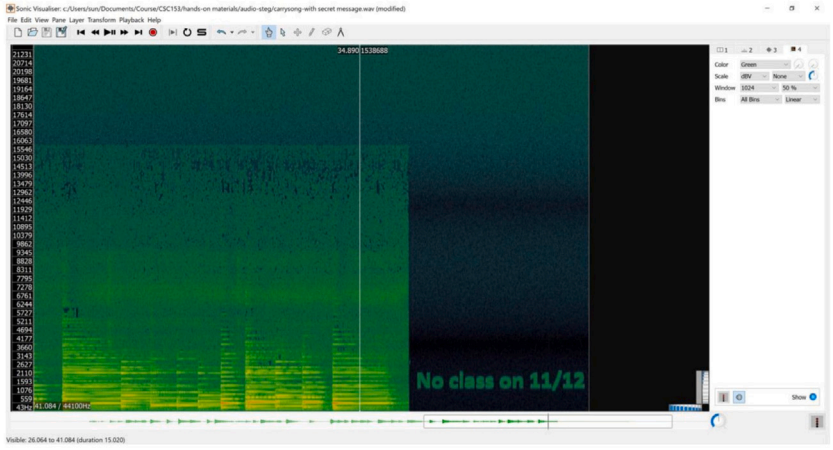


Fig. 1. An excerpt of step-by-step instructions from example INFER Lab: Steganography with Audio File.

on the labs following these instructions, requiring minimal help or guidance from the instructors. The instructions and lab materials needed (virtual machine images, artifact files, etc.) are available through the project website (INFER Project, 2026b) and GitHub (INFER Project, 2026a). All INFER labs, instructions, and environments are freely and openly available to any institution or individual worldwide, with no licensing, payment, or geographic restriction.

Each lab directory on the GitHub repository contains comprehensive materials, including lab instructions, relevant files, and tools for practice. For example, Fig. 2 highlights a repository focused on steganography. This repository includes a PDF file with lab instructions, input files such as "fun.bmp" and "lake.jpeg.cpt", and the tool "stegseek_06.1.deb" for practice. To reinforce learning, each lab concludes with review questions and optional bonus challenges. These challenges are designed to deepen the learner's understanding and encourage independent problem-solving. Although step-by-step instructions are provided for the primary tasks, the bonus challenges require learners to apply the knowledge and skills gained during the lab to complete the tasks independently.

3.4. Lab environment and deployment

To provide a comprehensive learning experience, the labs utilize CAINE, Windows, and Kali Linux operating systems (OS). This variety ensures that students are prepared to navigate different OS environments that they may encounter in professional practice. Windows, as one of the world's most widely used desktop operating systems, is a critical platform for digital forensic investigations. Familiarity with its essential tools and mechanisms is essential for effective forensic analysis. Kali Linux is an open-source distribution specifically designed for penetration testing, computer forensics, ethical hacking, and reverse engineering (Kali Linux, 2025). Similarly, CAINE (Computer Aided INvestigative Environment) is an open-source Linux distribution designed to support digital forensic investigations (CAINE, 2025). Both operating

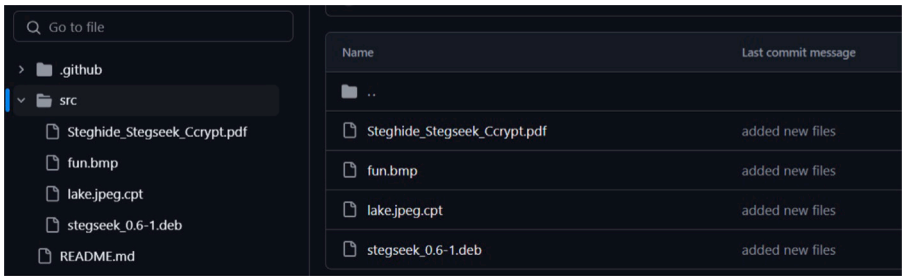


Fig. 2. Steganography hands-on repository.

systems come equipped with an extensive array of pre-installed forensic tools, enabling tasks such as disk imaging, analysis, and memory forensics. These tools provide the foundational capabilities required to collect and analyze digital evidence.

In addition to the pre-installed tools, the labs incorporate other free and readily accessible software such as WinPMem, Ghidra, and FTK Imager. For tools that require separate installation, we provide detailed installation links and setup instructions in the lab documentation.

Regarding hardware requirements, INFER labs are designed to run on a standard personal computer and require no specialized forensic equipment. Each lab runs within a single virtual machine using VirtualBox or VMware, with the lab materials mounted as an ISO image. The INFER lab ISOs are lightweight, ranging from approximately 1 MB to 500 MB. Most of the storage requirement comes from the operating system in the VM. The Linux-based environments, including CAINE and Kali, typically require approximately 20–30 GB, while the Windows-based labs may require up to 50–64 GB. In practice, a multi-core processor, at least 8 GB of RAM, and sufficient free disk space for the selected VM are adequate, although 16 GB of RAM is recommended when running forensic workloads. These requirements support INFER's portability and accessibility goals.

4. Evaluation

4.1. Study design and overview

To assess the impact and effectiveness of the INFER labs, we performed a multi-population study including three distinct population groups: approximately 50 college students across undergraduate and graduate courses — the majority of them are pursuing majors in computer science, cybersecurity, or related fields, about 20 K-12 teachers in a graduate summer DF course, and 11 college faculty in a two-day workshop. Due to the limited time allocated for labs within the courses and differences in audience, only a subset of INFER labs was used in each context. It is not feasible to incorporate all the designed labs into one course, as each course requires specific topics to be covered. For the student group, we were able to cover seven labs in a semester course. The K-12 teachers worked through four labs (Linux Acquisition, Linux Operation, Set Up VM, and Steganography), and the faculty worked through five labs (Linux Acquisition, Autopsy, Image Steganography, Audio Steganography, and Reverse Engineering). Although each group worked through a different subset, several core labs, such as Linux Acquisition, Set Up VM, and Steganography, were shared across multiple groups. Besides that, as mentioned above, it is not practical for a single course or workshop to cover all 18 labs, given that each audience group has distinct learning goals, background knowledge, and time constraints. The varying coverage and flexibility of the adopted combination, however, reflect INFER's intent as an adaptable resource. Educators can select suitable labs for their context and classrooms, rather than a fixed curriculum delivered uniformly.

In addition to these three primary study groups, INFER was later used in two further K-12 teacher cohorts: a second summer cohort (N = 15) and an advanced course (N = 28) that included returning teachers from earlier cohorts. The assessed lab and quiz outcomes of these two groups are reported in Section 4.4 (Table 4).

Our data collection and processing procedures for all study groups adhered to ethical guidelines for human subject research and have obtained approval from the Institutional Review Board (IRB). For each group, we conducted different surveys, including pre- and post-course surveys that evaluate participants' self-reported knowledge, interest, confidence, and career readiness, as well as surveys for each hands-on lab they worked on to evaluate the perceived clarity, difficulty, usefulness, and engagement of each lab.

Additionally, the pre-course confidence percentages in understanding and performing DF tasks for the teacher group reported in our prior work (Huynh et al., 2025a) included duplicate survey responses from some participants. In this extended work, duplicate submissions were identified and removed, retaining only one response per participant. Despite this correction, the overall findings remain consistent with the original conclusions.

4.2. Student learning outcomes

4.2.1. Student background and prior exposure

In the pre-course survey, students were asked to rate their interest in Digital Forensics on a scale from 1 (low) to 5 (high). We found that 98% of the students expressed interest in the topic. More than half of the students were considering or interested in pursuing a career related to digital forensics. However, most of the students were initially unfamiliar with digital forensics and perceived it as a complex subject. When asked about their confidence in their skills and career readiness in digital forensics, only 15 out of 51 students reported average or high confidence, with just 3 students selecting a rating of 4 or 5.

More than 60% of the students were unfamiliar with how to setup a Virtual Machine (VM) for a digital forensic workstation, a fundamental but critical step in forensics and many other cybersecurity tasks. Most undergraduate students were unable to perform this task, while 60% of graduate students were comfortable preparing a virtual machine environment before the course.

In all labs, we emphasize the importance of ensuring the integrity of the evidence data. Since the data and devices of evidence might contain malware, opening them on the host machine is risky. Additionally, data on the investigator's host machine could interfere with the evidence, compromising its integrity. Therefore, understanding how to set up a sandbox, such as the virtual machine environment, to handle potentially harmful data is essential in cybersecurity, as well as in digital forensics.

Only 15% of the students had experience with the acquisition of Linux or Windows data, the first and most crucial step in collecting and retrieving valid data. Unlike Network Forensics, which involves tools like Wireshark and is typically covered in Computer Network courses, most other topics in digital forensics were unfamiliar to the students.

In general, the confidence ratings for precourse skills were low across both groups (median = 1, IQR = 2). A Mann–Whitney U test comparing the baseline confidence of undergraduate and graduate students did not find significant differences between the two groups

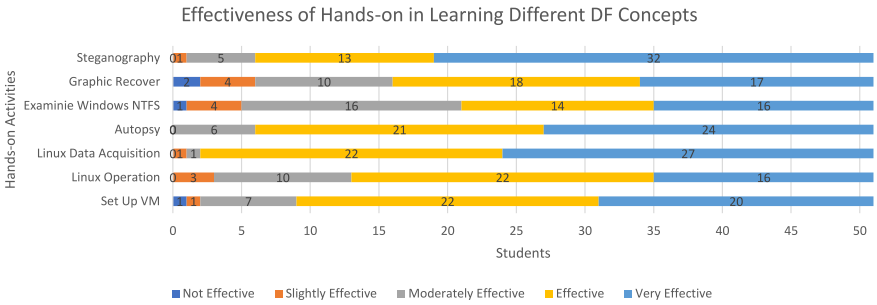


Fig. 3. Effectiveness of INFER labs for learning different DF concepts.

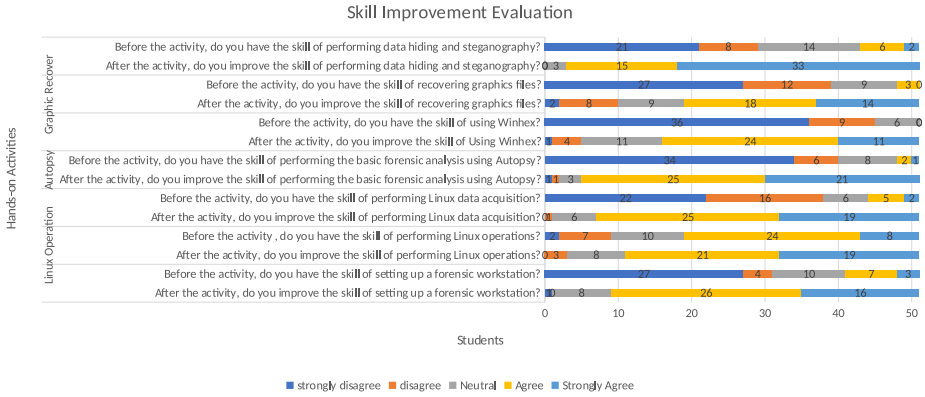


Fig. 4. Students' skill improvement through INFER labs.

(U = 257, p = 0.18). This result suggests that most students, regardless of levels, entered the course with low confidence and similarly limited prior experience in Digital Forensics.

4.2.2. Learning experience across the INFER labs

In the surveys for individual hands-on activities, we gathered student feedback through 5-point Likert rating scale questions and short-answer responses to assess the labs' impact on their learning in Digital Forensics. For example, we ask questions such as: *How **difficult** did you find this activity? How do you feel about the **clarity** of the activity instruction? Do you feel Activity 6 was **effective** in helping you learn digital forensics concepts such as how to use Winhex to recover graphics files? Does Activity 6 make you **interested** in the topic of using Winhex to recover graphic files?*

As shown in Fig. 3, when students were asked if the hands-on labs helped them effectively learn digital forensic concepts, most of the participants found the activities helpful. However, there were often students who ran into technical challenges in some labs. For example, some students encountered challenges during the "Setup VM" task, particularly when setting up the virtual environment on Mac machines. Details of technical challenges encountered by some labs are further discussed in Section 5.

The improvement in students' skills before and after participating in the activities is significant, as illustrated in Fig. 4. For instance, in the "Examine Windows NTFS" activity, none of the students initially possessed the skill to use WinHex. However, after completing the hands-on lab, this number increased to 35 students, reflecting an improvement of almost 70%. Similarly, activities such as "Steganography" and "Set Up VM" demonstrated nearly 100% skill enhancement rates. Upon examining the cases where students reported no improvement in their skills, their short-answer feedback revealed that these instances were mainly due to technical issues with Mac machines, as explained in more detail in Section 5.

Fig. 5 summarizes the median ratings in all seven labs on difficulty, clarity of instructions, effectiveness in learning the material, and whether the lab increased students' perceived interest in the topic. In general, students rated the labs positively across all dimensions, with most labs scoring at or above the neutral midpoint. The INFER hands-on activities significantly increased students' interest in digital forensics and effectively enhanced their skills across various topics and tasks. "Steganography", "Data Acquisition", and the real-world case-based "Autopsy" lab garnered the highest interest ratings. However, activities such as "Windows NTFS" and "Graphic Recovery" required extensive understanding and interaction with hex data files, which proved to be more challenging for many undergraduate students. Consequently, these topics received lower interest ratings among this group. However, the inspiration rates among graduate students for these two topics were higher, at 90% and 70% respectively, indicating that prior technical background might change how students engage with more demanding content. This aligns with the difficulty and clarity ratings for these labs discussed above.

When comparing the undergraduate and graduate groups, the ratings were largely consistent across both levels, with the only exception being effectiveness for Linux Acquisition, where graduate students rated it slightly higher (median 5 vs 4; U = 109, p = 0.01). This difference may stem from graduate students being more familiar with Linux environments from prior coursework or professional experience. Besides that, this activity is one of the more technical labs. Therefore, students who already have some background in using the command line, even if that prior experience did not fully show in the baseline confidence rating, will find it more effective and meaningful.

To examine if the instruction clarity relates to the lab difficulty, we used Spearman's rank correlation (ρ), which measures the strength of the relationship between two variables, where ρ is closer to 1, meaning a stronger positive association. As shown in Fig. 6, instruction clarity was significantly correlated with perceived ease in six of the seven labs ($\rho \in [0.32, 0.56], p < .05$). This result confirms that clearer instructions consistently made the labs more approachable for students. The exception case was the Steganography lab ($\rho = 0.07$, not significant), which had low activity difficulty regardless of the instruction clarity rating.

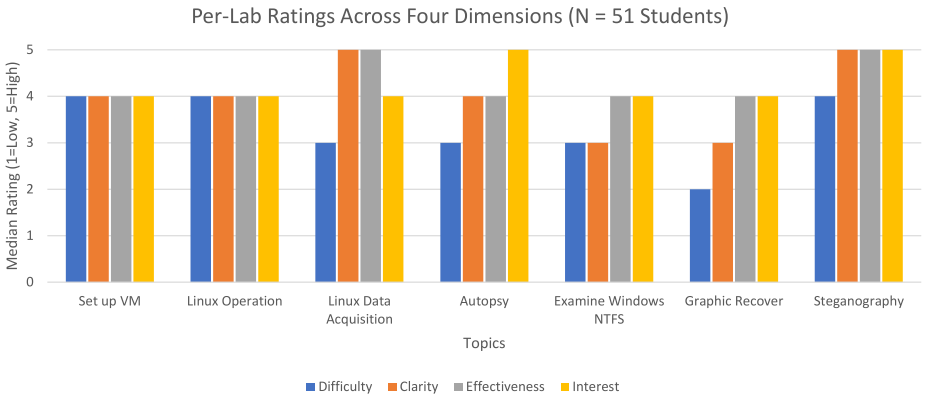


Fig. 5. INFER per-lab rating across difficulty, clarity, effectiveness, and interest.

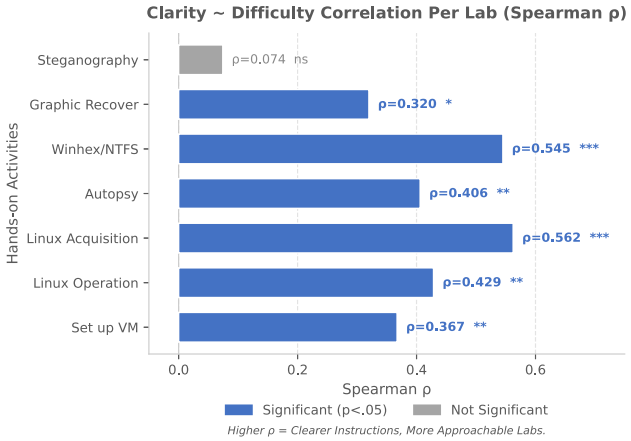


Fig. 6. Instruction clarity and difficulty correlation.

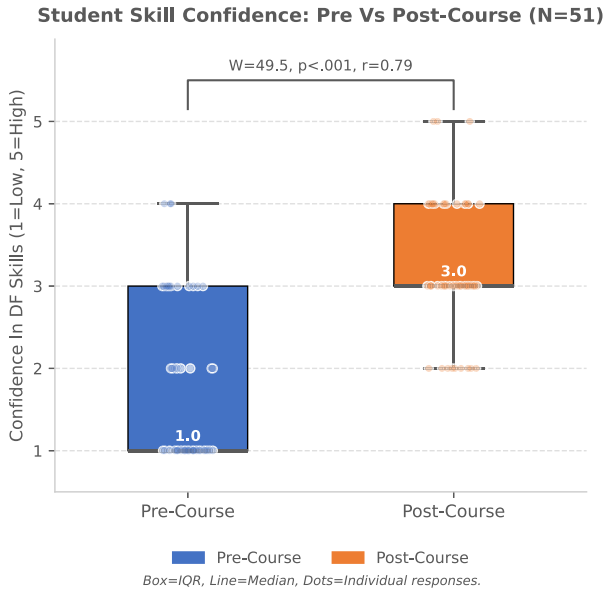


Fig. 7. Students' confidence in DF skill and career readiness before vs. after the course.

This is not a design limitation but rather a matter of the topic itself. Steganography tools such as steghide and stegseek provide straightforward, user-friendly interfaces that allow students to immediately see the results without requiring a deep technical background. We also find that clarity correlated with effectiveness across most labs (significant in 5 out of 7 labs, $\rho \in [0.31, 0.75], p < .05$), and interest level correlated with effectiveness (significant in 5 out of 7 labs, $\rho \in [0.42, 0.7], p < .05$), indicating that the instruction quality and student engagement are together strengthening the overall learning experience.

4.2.3. Skill and confidence growth

After completing the course, students also completed a post-course survey. Since the confidence ratings are ordinal and on a 5-point Likert scale, we applied the Wilcoxon signed-rank test to compare the pre- and post-course ratings. The comparison is shown in Fig. 7, which shows the distribution of students' skill confidence before and after the course. The box spans the interquartile range (IQR), which displays where most of the responses fell, specifically the middle 50% of responses. The horizontal white line within the box represents the median, and the individual dots represent each student's actual response. As shown in Fig. 7, there was a significant increase in the number of students who somewhat agreed or strongly agreed that their ability to understand and perform digital forensic tasks had improved. The median of students' skill confidence increased from 1 to 3 by the end of the course ($W = 49.5$, $p < 0.001$, $r = 0.79$), showing a large effect size. Nearly all students reported being able to set up their own VMs for forensic investigations and perform data acquisition in both Windows and Linux, compared to just 5%–32% at the start of the course (Fig. 8).

For advanced topics such as "Android Forensics" and "Drone Forensics", none of the students had prior experience with the material before the course. This is in part due to the fact that many of the tools involved are not free and not everyone has access to expensive smartphone or drone technology.

Additionally, we found that labs focusing on binary analysis or assembly code, such as "Reverse Engineering", as well as labs requiring a combination of various digital forensics skills and knowledge, such as "Social Network Forensics", "Memory Forensics", and "Drone Forensics", proved to be more challenging for students. As a result, the understanding and confidence levels for these exercises were generally lower compared to those for other labs.

However, the feedback that we received from the students after completing the hands-on tasks was largely positive. Below are some of

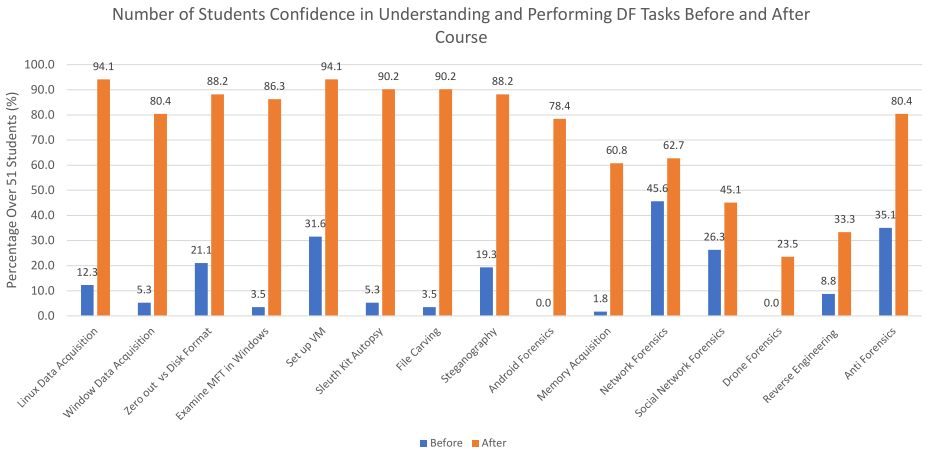


Fig. 8. Number of students confident in performing DF tasks before vs. after the course.

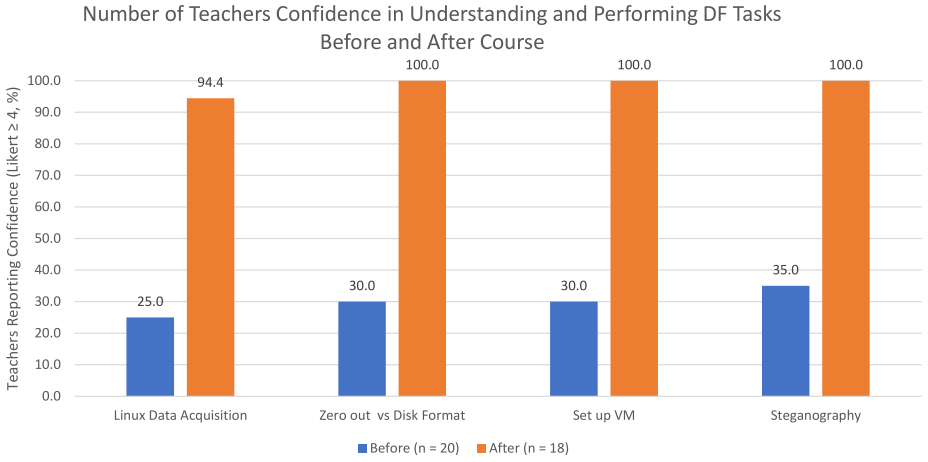


Fig. 9. Teachers' confidence level in performing DF tasks before vs. after the course.

the comments from students following the course and their hands-on lab experiences:

"It is a truly enriching experience to learn more about digital forensics through hands-on activities with different tools, such as how to recover deleted or hidden data and bypass security measures on Android".

"Much more interested now than I was before. May choose this as my career path. Thanks for the introduction!"

"This was a phenomenal experience. I look forward to implementing what I've learned in the future. I look to brush up on a few more concepts and skills before applying them to a job. This class is a wonderful stepping stone".

"Overall, a great and engaging class. Really enjoyed the activities and gained a greater knowledge in digital forensics".

"Very interesting and allows you to explore different areas of forensics".

4.3. Educator perspective

4.3.1. K-12 teacher outcome

The graduate-level summer course in digital forensics has approximately 20 teachers from K-12 schools (mostly high schools) nationwide. All participants were teaching or planning to teach courses in cybersecurity or other courses related to computer science, engineering, or similar disciplines.

Before participating in hands-on labs, most teachers had little or no familiarity with digital forensics, with only about 30% having prior experience or confidence in the subject. Considering that most teachers were teaching at the high school level, the course used only a small subset of INFER labs as hands-on activities: "Linux Data Acquisition", "Linux Operation", "Setting Up Virtual Machines for digital forensics investigations" and "Steganography". The feedback collected from the teachers on confidence is presented in Fig. 9 (updated results after removing duplicate survey responses). Of the 20 teachers who took the pre-course survey, 18 of them completed the post-course survey. The results shown in Figs. 9 and 10 are based on these 18 responses. Although not covering all topics, INFER-based hands-on lab practice significantly improved participants' confidence, with 100% reporting confidence in performing digital forensic tasks by the end of the course. Wilcoxon signed-rank tests also show gains for every lab (the median after the course is 4-5, from 1-2 before taking the activities). "Set up VM" and "Steganography" achieve the highest gains, with every teacher reporting higher confidence after completing the INFER labs. This indicates that the teachers gained sufficient technical background to teach their students in these topics confidently.

Even though both Computer Science (CS) Students and K-12 teachers experienced the same labs with the same instructions, they perceived the difficulty differently. Here, we performed a Mann–Whitney U test to compare the perspectives of these two independent groups. The box plots show the distributions of the difficulty ratings. The higher value means the participant found it easier. The box spans the interquartile range; the horizontal line shows the median; and individual dots represent actual rating responses. In cases where no box appears, such as Linux Operation for K-12 teachers, most respondents rated the difficulty at the same value (median = 3). Hence, the interquartile

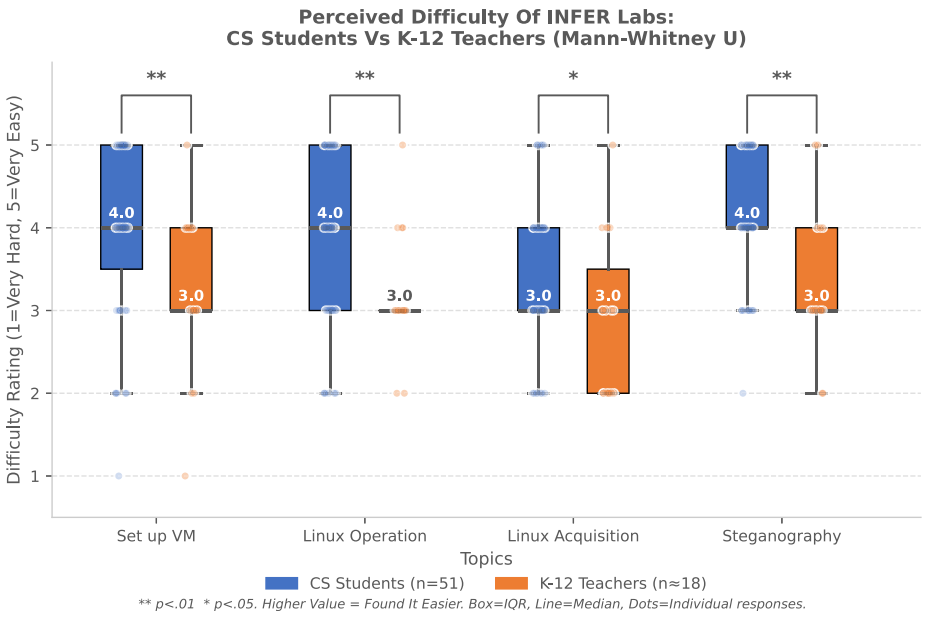


Fig. 10. Perceived difficulty of INFER Labs: CS Students vs K-12 Teachers.

range is 0, and no visible box in the plot. From Fig. 10, the students' groups rated most labs significantly easier than the teachers' groups ($p < 0.01$); only in the "Linux Acquisition" Lab did both groups rate it the same difficulty level (median = 3). The existence of this gap is expected, given the differences in background and perspective between these two groups. While K-12 teachers are comfortable with programming, networking, and general CS problems, the K-12 CS curriculum does not typically cover hex values, binary file structures, or raw disk data. On the other hand, college CS courses usually expose students to low-level computing concepts, more Linux command work, and even assembly or system programming. Besides that, when rating the difficulty, teachers may assess it not only from their own experience but also through their students' lens. They might think, "Can my students do this?" instead of just "Can I do this?" Hence, this perspective from the educators will affect how they rate the difficulty of the hands-on labs. INFER already partially addresses this through detailed step-by-step instructions, case-based scenarios, and modular structures. To further support non-CS audiences, supplementary online tools (hex and binary converters, etc.) and resources could help bridge the remaining gap before teachers engage with the full lab. The core principle here is helping teachers fully understand the material before they can confidently deliver it to their students. Furthermore, teachers are encouraged to not only use the labs as provided but also adapt and integrate them into their own teaching plans and curriculum — a process that is discussed further in Section 4.3.3.

4.3.2. Faculty workshop

A two-day development workshop was also held for faculty from community colleges and four-year universities to disseminate the designed labs. The event brought together 11 faculty members and professors from various institutions in the United States. We provided an introduction to the INFER project and training on how to use labs in digital forensics courses.

Participants also received a detailed guided walkthrough of five hands-on labs during the workshop, and then were given the opportunity to work on the labs independently, taking the students' role. These five labs, including "Linux Data Acquisition with Zero Out and Disk Formatting", "Sleuth Kit Autopsy", "Image Steganography", "Audio Steganography", and "Reverse Engineering", are examples demonstrating how the faculty can use them in their own class. With these examples, they also know how to access and use the materials for the remaining labs. At the end of the workshop, participant surveys were conducted to obtain the faculty's opinions on INFER labs' design and effectiveness. All participants (100%) found the INFER labs valuable and would recommend them to colleagues or friends teaching cybersecurity or digital forensics-related courses. Many of them have plans to adopt INFER labs in their own courses or start a digital forensics course in their institution.

Below are reflections and feedback from some teachers and professors who participated in the course and workshop:

"The synchronous instructions coupled with the hands-on labs offered the best of both worlds".

"The hands-on labs are really useful in my teaching and research".

"I really liked the accessibility of the labs and being able to continue to work with them outside of the workshop".

"We would like to use the labs in our digital forensics courses".

"The INFER labs offer multiple opportunities for advancing digital forensics (DF) research and education. We can create new courses for undergraduate and graduate students, integrating selected DF labs into these new courses as well as existing ones. Additionally, we can conduct DF research to develop new techniques that enhance current DF methodologies and skills".

4.3.3. From training to teaching: INFER labs and classroom integration

As mentioned earlier, one of the main design principles of INFER labs is the ability for adoption. We evaluate this aspect through two sources: video demonstration and discussion board submissions from a final project in a subsequent advanced course, and an anonymous follow-up survey sent to both faculty workshop participants and K-12 teachers who had taken the course, from which we received 17 responses (14 K-12 teachers and 3 university faculty).

Beyond the self-reported confidence and satisfaction discussed above, classroom adoption provides additional evidence of INFER's practical value in real instructional settings. As shown in Table 2, several educators from both K-12 and University faculty who participated in the training or workshop later integrated INFER labs into their own courses and classrooms. These results suggest that the labs were not only rated positively in surveys but also considered usable, sufficiently practical, and valuable for deployment by different educators. However, this evidence should be interpreted as support for INFER's adaptability, usability, and educator-perceived value, rather than as a

Table 2
Adoption summary: Educators who used INFER labs.

Role	Labs used	Students	Deployed	Difficulty	Cont.[a]
Teacher	Data Hiding, Acquisition, Autopsy, NTFS	21–50	Project-based	Easy	4
Teacher	Steganography (Image)	10–20	Demo	Moderate	3
Teacher	VM Setup, Autopsy, Social Network Forensics, Audio & Image Steg	10–20	Demo, Class activities	Moderate	4
Teacher	Data Hiding, NTFS, Steganography (Image)	10–20	Demo, Slides	Moderate	4
Teacher & Faculty	Data Hiding, Autopsy	10–20	Investigation	Moderate	5
Faculty	Reverse Engineering (Ghidra, IDA)	50+	Lecture notes	Moderate	5
Faculty	Steganography (Image)	21–50	Lab Project	Easy	5

[a] *Note.* Continue using INFER for future courses: 1 = very unlikely, 5 = very likely.

Table 3
Median effectiveness ratings from educators who used INFER labs.

Dimension	Median (1–5)	n
Portable environment reduced setup time	4	7
Able to adjust labs to fit course	4	7
Students able to complete labs	4	6
Improved understanding of DF concepts	5	6
Increased engagement in cybersecurity	5	6
Improved practical DF skills	5	6
Students used forensic tools successfully	4	6
Helped understand real-world DF processes	4	6
Likelihood to continue using INFER	4	7

Note. Scale: 1 = strongly disagree/very unlikely, 5 = strongly agree/very likely.

direct measure of students' outcomes of those educators' own students, which remains beyond the scope of this study.

For the final course project, K-12 teachers could choose either of the following directions: performing a deep study on one area of DF beyond what was covered in the course or transferring what they learned in the course to their own curriculum for their classes. The first direction focuses more on technical depth and tool exploration, while the second focuses more on instructional and design. The approaches to this project are very creative and diverse. There are standalone intro lessons, multi-day unit lessons, live in-class demos, project-based lessons, or full lesson plans with slide decks and student guides. Some notable creative adaptations beyond the INFER hands-on include combining CTFs (picoCTF, cyber.org) with INFER labs or building a steganography website for students to investigate. Besides designing their own lectures, teachers actively engaged with each other's work through discussion boards, such as sharing adapted plans, requesting INFER materials, and using each other's labs in their own classrooms. These observations indicate that INFER has the potential to encourage and inspire a broader community of practice in Digital Forensics education.

Of the 17 responses received from the follow-up survey, seven educators confirmed using INFER labs in their actual classes. We summarized their responses in Table 2, with each row representing one response. As shown in the table, the deployment methods range from lecture notes, demos, and slides to hands-on investigation, assignments, and lab projects. "Steganography" was the most widely used lab, followed by "Autopsy" and "Data Hiding". Class sizes range from 10 to 50 students per educator, with one university faculty member reaching over 50 students through Reverse Engineering labs lecture. The INFER implementation difficulty was rated mostly Moderate or Easy, and preparation time was usually less than an hour, except for VM setup (2–3 h). This is because of the extra time taken to install an Operating System like Caine or Kali on a VM. Most educators were also likely to continue using INFER labs in their future courses.

To evaluate the effectiveness and impact of the hands-on labs on their students, we asked educators to rate various aspects, such as setup time, student improvement, and course fit, on a 5-point Likert scale. Note that, for some questions, we received only 6 out of 7 ratings, since one of the faculty members who used the "Reverse Engineering" labs delivered the hands-on as lecture notes rather than a project-based approach. Therefore, some questions about student tool use and skill improvement may not have been applicable. As shown in Table 3, the ratings are consistently positive across all aspects.

For the remaining educators who have not used the labs in actual class, the most common reason was simply that they had not yet taught a relevant course, followed by a lack of time to integrate them into their own curriculum. Some educators, especially K-12 teachers, had difficulty with integration because of the infrastructure constraints, such as Chromebook-only environments and restricted software installation policies from their school. We discuss a cloud-based deployment model to address this barrier in Section 5. Despite not having used the lab yet, 8 out of 10 educators indicated that they plan or are considering using INFER labs in the future. The others are either not teaching relevant courses or are constrained by Chromebook hardware limitations. This challenge is not unique to INFER but is worth addressing through alternative approaches, for example, web-based or cloud environments, which the community could consider as a future direction.

4.4. Instructor-assessed outcomes

To complement the survey-based evaluation, we also examined available course outcomes from different participant groups where records were accessible. In every course that used INFER, hands-on lab work was graded as a requirement for course completion. There are also quizzes along the course and end-of-lab questions in several labs that test understanding of the underlying tools, processes, and knowledge, rather than just tool operation. Since INFER was used across different instructional settings, the grading instruments varied across cohorts. Each course had its own audience, duration, goals, rubric, and selected lab subsets. Therefore, these scores serve as objective, instructor-assessed evidence that participants were able to complete the assigned forensics tasks, rather than as direct cross-population comparisons or standardized pre- and post-learning gains. Table 4 summarizes the median graded outcomes for different participant groups where such records exist.

The graded records show that participants were generally able to complete the assigned forensic tasks and answer the knowledge-check quizzes after using INFER. These graded outcomes also show a clear ceiling effect with medians ranging from 96 to 99 and narrow interquartile ranges. Since there was no pre-course baseline, these results reflect task performance rather than measured learning gains. For the first K-12 cohort, complete quiz and hands-on lab records

Table 4
Instructor-assessed learning outcomes.

Participants	N	Labs used	Labs (Max 100), median (IQR)	Quiz (Max 100), median (IQR)
Graduate students	10	Set up VM, Linux Operation, Linux Data Acquisition, Autopsy, Examine Windows NTFS, Recover Graphic File, Steganography	96.6 (93.4–99.7)	–
K-12 teachers (first cohort, reported in paper)	19	Set up VM, Linux Operation, Linux Data Acquisition, Steganography	96.8 (95.0–97.5)	98.6 (97.7–99.4)
K-12 teachers (second cohort)	15	Set up VM, Linux Operation, Linux Data Acquisition, Steganography	98.8 (98.8–100)	99.1 (94.9–99.8)
K-12 teachers (advanced course)[a]	28	Autopsy, Examine Windows NTFS, Forensics on Discord	98.7 (90.0–100)	96.4 (93.1–98.6)

[a] *Note.* The advanced cohort included returning K-12 teachers from prior courses.

were available for 19 teachers, a count that differs from the survey-based totals reported above because these lab and quiz outcomes were tracked separately from survey participation. Of the 50 college students, 10 were graduates, and the remaining were undergraduates. Thus, the graduate row in Table 4 (N = 10) covers the full graduate group. These records cover only active participants, and graded data for the undergraduate student group are no longer available. Therefore, the instructor-assessed outcomes should be interpreted as complementary validation of the self-reported results rather than a primary, standardized, or comparative controlled measure of learning gain.

5. Limitations and future work

Keeping software up to date is essential given the rapid evolution of digital forensic technologies. For example, Android, IoT, and drone forensic software evolves quickly, making some tools used in the labs potentially obsolete over time. Continuous effort is therefore required to update the lab environments and instructional materials accordingly. Additionally, some software packages, such as Belkasoft X and WinHex, provide only limited-time free access or restrict advanced functionality in their free editions, which may affect long-term accessibility. We intentionally included these tools because one of INFER's goals is to expose learners to software commonly used in professional forensic practice. At the same time, long-term sustainability does not depend on any single tool. Several labs already use mature open-source alternatives, such as Autopsy and The Sleuth Kit, and the learning objectives are generally tool-independent. Because INFER is fully open source, educators and the community can readily replace commercial or discontinued software with suitable open-source or newer alternatives as the forensic software ecosystem evolves, thereby supporting the long-term sustainability of the platform.

To enhance accessibility, we are working to extend lab instructions to Mac environments, particularly for preparing virtual machines and images, as several tools are not Mac-compatible and require alternative setup procedures. Due to Apple's proprietary silicon chips, Caine VM image, which only supports x86_64 architecture, are incompatible with Mac's unique architecture. We recognize this issue and have worked to develop additional instructions for this special case. For Mac machines, alternative software and setup steps may be required. For example, VMware Fusion or Workstation can be used on Mac as alternatives to VirtualBox, which is no longer supported on mac OS. However, VMs must be compatible with Mac processors (i.e., ARM architecture) to run on Mac, even when using VMware. For VM images incompatible with Mac processors, the QEMU emulator (QEMU, 2025) provides an effective solution. QEMU is a free, open-source software that is easy to install and can be managed efficiently through the command line in the terminal. We also clarify that the architecture constraint is narrower than it may initially appear. Among the operating systems used in INFER labs, only CAINE and Santoku (used in Lab 4 — "Linux Workstation Setup and Analysis with Autopsy" and Lab 11 — "Android Forensics") are limited to x86_64. Windows and Kali Linux, which support most of INFER hands-on exercises, both provide ARM-compatible images and can run natively on Apple Silicon. Lab 4 was intentionally designed with Caine to expose learners to this widely used Linux distribution for digital forensics. In other Linux labs, Caine is just an interchangeable alternative to Kali. Therefore, students with ARM-based machines can use the provided Kali image instead. The only other x86_64 dependency is Santoku, which is used in the "Android Forensics" lab. For these two cases, the QEMU-based emulation path described above provides a workable solution on ARM architecture hardware. Thus, learners on Mac or other ARM platforms can complete most INFER labs natively, with emulation required only for these two specialized exercises.

Cloud-based deployment is a promising direction for lowering adoption barriers, particularly for institutions with restrictive software installation policies or Chromebook-only environments, as reported by some K-12 educators in Section 4.3.3. In principle, this model could be implemented through browser-accessible VM or centrally hosted containerized lab environments, allowing learners to access and complete the labs without installing the required OS or software locally. However, this approach also introduces additional requirements, including persistent hosting infrastructure, ongoing maintenance, and recurring operational costs. Since INFER is open-source, such environments could also be hosted or adapted by the community and adopting institutions rather than being centrally operated. Therefore, we consider cloud-based hosting a valuable direction for future work, while the portable VM and QEMU-based approaches discussed above remain the current deployment options for accessibility.

In addition to the technical infrastructure limitations in some institutions and the technical challenges mentioned above, several limitations and factors constrain the generalizability of our findings and suggest directions for future work. First, the evaluation relies on self-reported survey data, and outcomes were measured during and immediately after the course. While these reports capture learners' perceptions and engagement, they do not directly measure objective competence, nor do they address long-term retention. Therefore, future longitudinal studies could be used to assess learners' long-term retention. Besides that, the instructor-assessed lab and quiz outcomes in Section 4.4 that complement these self-reported measures are themselves incomplete, as graded records for the undergraduate student group are no longer available and cover only a subset of participants. Second, because no single course or workshop could accommodate all 18 labs, cross-population comparison was possible only for a small set of foundational labs; comparative evaluation of the more advanced labs remains future work. Another constraint the study has not addressed is the need for

Table 5
List of INFER topics and hands-on labs.

Topic & Hands-on Labs	Key skills (NICE)	Key knowledge (NICE)
Data Acquisition H1: Data acquisition using dd/dcfldd H2: Windows acquisition tools	Live **(S0935)** and Deadbox acquisition **(S0936)**; Preserving evidence integrity **(S0589)**	Evidence seizure **(K0795)**; Evidence preservation **(K0796)**; Evidence handling **(K1153)**
Windows Forensics H3: Examine Windows NTFS using *WinHex*	File system forensic analysis **(S0860)**; Digital forensic analysis **(S0612)**	File systems **(K0794)**; System file characteristics **(K0808)**; Forensic data characteristics **(K0809)**
Linux Forensics H4: Linux workstation setup and analysis with *Autopsy*	Digital evidence analysis **(S0856)**; Processing digital evidence **(S0608)**	Forensic tools **(K1092)**; OS systems **(K0744)**; OS internals **(K0923)**
File Carving H5: File carving H6: Recover graphic file	Identifying **(S0603)** and Extracting **(S0604)** forensics data	Data carving tools **(K0850)**; Persistent data **(K0804)**; Data bytes **(K1281)**
Data Hiding H7: Basic data hiding and steganography H8: Steganography with audio H9: Steganography with image H10: Develop an image steganography tool	Decrypting information **(S0378)**; Critical thinking **(S0431)**	Steganography **(K1220)**; Data concealment **(K0941)**; Forensic data principles **(K0696)**
Mobile Forensics H11: Android forensics	Collecting digital evidence **(S0607)**; Data inspection **(S0937)**	Evidence extraction tools **(K1152)**; Communication systems **(K0812)**
Memory Forensics H12: Memory forensics	Memory dump analysis **(S0599)**; Volatile data analysis **(S0624)**	OS internals **(K0923)**; Forensic data characteristics **(K0809)**; Forensic data principles **(K0696)**
Network Forensics H13: Network forensics with *Xplico*	Packet analysis **(S0156)**; Network traffic packet analysis **(S0875)**;	Network protocols **(K0674)**; Networking principles **(K0983)**; Network monitoring tools **(K1175)**
Social Network Forensics H14: Forensics on Discord	Intelligence collection analysis **(S0499)**; Data analysis **(S0854)**	Information searching tools **(K0979)**; Communication systems **(K0812)**; Data types **(K1281)**
CPS/IoT Forensics H15: Drone forensics	Collecting digital evidence **(S0607)**; Data analysis **(S0854)**	Physical computer components **(K0786)**; Communication systems **(K0812)**; Computer engineering principles **(K0947)**
Reverse Engineering H16: Reverse engineering with IDA H17: Reverse engineering with Ghidra	Source code analysis **(S0623)**; Interpreting debugger results **(S0625)**; Dynamic analysis **(S0857)**	Reverse engineering principles **(K0851)**; Binary analysis tools **(K0914)**; Debugging tools **(K0855)**
Anti-Forensics H18: Anti-forensics techniques	Decrypting information **(S0378)**; Critical thinking **(S0431)**	Anti-forensics tools **(K0852)**; Code obfuscation **(K1016)**; File extensions **(K0793)**

evaluation with more diverse audiences, especially in non-academic settings (e.g., workforce development). Broader, multi-institutional, and population-based studies would strengthen generalizability. Relatedly, all INFER materials are currently available only in English, which may limit accessibility for non-English-speaking learners and institutions. Translating the labs could be a valuable future direction for broadening the resource's reach.

For the content development, we aim to integrate more real-world attack scenarios and case studies, as suggested by Xu et al. (2022), Wang et al. (2019). This will provide students with a more practical, engaging, and effective learning experience. We will periodically review and update the tools and software to ensure they remain up to date and accessible.

6. Conclusion

The demand for a highly skilled cybersecurity workforce, particularly in digital forensics, continues to grow. INFER Labs aims to bridge this gap by offering a practical, interactive, comprehensive, and accessible resource for digital forensic education. Survey results from students, teachers, and college faculty provided very positive feedback and documented classroom adoption by independent educators, further reinforcing the project's usability, perceived effectiveness, and potential to enhance digital forensics training. As INFER is free and open-source, future collaborations and contributions can further expand and refine the materials while keeping them practical and aligned with real-world applications.

CRediT authorship contribution statement

Tran Ngoc Bao Huynh: Writing – review & editing, Writing – original draft, Visualization, Software. **Haowen Xu:** Writing – review & editing, Writing – original draft. **Brian Almaguer:** Writing – review & editing, Writing – original draft, Software. **Jun Dai:** Writing – review & editing, Supervision, Funding acquisition. **Xiaoyan Sun:** Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix 7

See Table 5.

Data availability

Research Link Provided

Ready-to-Use Hands-on Labs with Portable Operating Environments fo r Digital Forensics Education (Original data) (INFER)

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